Filter-Induced and Learned Deadlocks in Safety-Filtered Reinforcement Learning for Mobile Robot Navigation

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Abstract

A control-barrier-function (CBF) safety filter can make a learned navigation policy collision-free, but the robot may then stall in front of an obstacle and time out. We analyse such deadlocks for a unicycle robot whose speed is limited by a one-step discrete-time CBF filter. We first characterise the filter geometrically. Its one-step safety guarantee does not depend on the commanded turn rate, so any modification of the turn rate preserves the guarantee. Near the keep-out boundary, the set of headings that permit motion shrinks to the half-plane that points away from the obstacle. We then classify the timeouts of ten proximal-policy-optimisation (PPO) agents and two classical controllers on 300 seeded test scenarios. In every scenario, a geometric oracle confirms that a path keeping the safety margin exists. Two mechanisms dominate, and they depend on how the agent was trained. Agents trained without the filter and run with it are *blocked by the filter* at the boundary (141 of 175 expert timeouts). Agents trained with the filter mostly *stop by themselves* before the boundary (122 of 195), so the filter is not the immediate cause. In paired comparisons on the expert stage, training with the filter lowered the filter-blocked rate by 14.8 percentage points (95 % interval [5.6, 24.4]) and raised the self-stop rate by 17.6 points [8.2, 27.0]. Sampling actions instead of using the policy mean removed most self-stops but no filter-blocked stalls. All results are from simulation with exact state and static circular obstacles, and they do not establish real-world safety. **[EVIDENCE REQUIRED:** Results of the correction-penalty baseline (E1) and of a heading-level remedy (E2) must be added before submission; see minimum_experiments.md.]

Keywords: safe reinforcement learning; control barrier functions; safety filter; deadlock; mobile robot navigation; liveness.

1 Introduction

Safety filters are a common way to make learned controllers usable in safety-critical settings [1, 2]. The learned policy proposes an action, and a filter, often a control barrier function (CBF) quadratic program [3, 4], replaces it with the closest action that keeps the system in a safe set. In robot navigation, the safe set excludes a margin around every obstacle, and the filter guarantees under its modelling assumptions that the robot never enters that margin.

The guarantee says nothing about reaching the goal. For model-based nominal controllers it is known that CBF filters can create undesired stable equilibria on the boundary of the safe set [5, 6], and multi-robot CBF controllers can deadlock [7–9]. When the nominal controller is a reinforcement-learning (RL) policy, the situation is less clear. The policy may have been trained with the filter in the loop [10–12], it reacts to observations rather than to the true state, and its behaviour near obstacles is learned rather than designed. Recent work shows that training with a sufficiently permissive filter loses no asymptotic performance [13], and that adding a penalty on the

filter's correction improves the performance of filter-trained agents in trajectory tracking [10]. To our knowledge, no published study characterises how learned navigation policies deadlock behind a filter, or how the training setup changes the deadlock mechanism. This statement rests on a targeted literature search; a systematic search is listed as a remaining gap.

This paper studies that question in a deliberately simple and fully reproducible setting. The robot is a unicycle, the obstacles are static circles, and the filter is a one-step discrete-time CBF on the speed [14]. The setting is simple enough that the filter's behaviour can be analysed exactly, and that every timeout can be checked against a geometric feasibility oracle.

Research questions.

RQ1 Which properties of a one-step discrete-time CBF speed filter determine where and how a filtered robot can stall?

RQ2 By which mechanisms do learned policies deadlock behind such a filter, and does training with the filter change the mechanism?

RQ3 Which simple remedies (stochastic deployment, the published correction penalty, heading-level modification) remove which mechanism? **[EVIDENCE REQUIRED:** Only stochastic deployment has been evaluated; E1 and E2 are required.]

Contributions.

1. A geometric characterisation of the one-step speed filter for the unicycle (Section 4). One-step safety is independent of the commanded turn rate (Proposition 1), so heading-level remedies can be added without weakening the guarantee. The set of headings that permit motion collapses as the barrier approaches zero (Proposition 2), and a stall bound follows (Corollary 1). Proposition 3 shows that self-stops cannot be removed by any change of the speed limit.

2. An evaluation protocol for deadlocks in filtered navigation. It combines a geometric feasibility oracle, a stall taxonomy (filter-blocked vs. self-stop) and uncertainty estimates over both training seeds and test scenarios (Section 5).

3. Empirical evidence from ten PPO agents and two classical controllers (Section 6). The two stall mechanisms exist, they depend on the training setup, they respond differently to stochastic deployment, and the filter-blocked stalls match the geometry predicted by Proposition 2.

We do not claim a new safety filter or a new training algorithm. The safety proof of the filter is standard [14] and is restated only for completeness.

2 Related work

Safety filters and CBFs. CBF quadratic programs [3, 4] and their discrete-time form [14, 15] are the standard tools for enforcing state constraints around arbitrary nominal controllers. High-order CBFs handle constraints with higher relative degree [16]. Predictive and reachability-based filters offer less conservative alternatives at higher computational cost [1, 17]. Robust, measurement-robust and stochastic variants address model and state uncertainty [18–21].

Table 1: Positioning. Y = yes; – = no; n.r. = not reported in the parts read; (Y) = partly.

Work	Learned policy	Filter in training	Navigation	Deadlock / liveness	Equilibria of learned policy	Feasibility oracle
Reis et al. [5]; Tan & Dimarogonas [6]	–	–	(Y)	Y	–	–
Grover et al. [7]; Jankovic et al. [8]; Chandra et al. [9]	–	–	Y	Y	–	n.r.
Krasowski et al. [12]	Y	Y	–	n.r.	–	–
Thumm et al. [31]	Y	Y	(Y)	n.r.	–	n.r.
Pizarro Bejarano et al. [10]	Y	Y	–	chattering	–	–
Oh et al. [13]	Y	Y	n.r.	n.r.	–	–
Yang et al. [11]	Y	Y	Y	n.r.	n.r.	n.r.
Marzari et al. [33]; Guo et al. [35]	Y	(Y)	Y	–	–	–
This work	Y	Y	Y	Y	Y	Y

Undesired equilibria and deadlock. Reis et al. [5] showed that CBF-QPs can introduce asymptotically stable equilibria on the safe-set boundary, and Tan and Dimarogonas [6] characterised them in general. In multi-robot navigation, deadlock was analysed as a force equilibrium on the boundary [7, 22] and resolved through instability of the deadlock set [8] or with discrete-time CBFs [9]. All of these analyses assume a model-based nominal controller. Potential fields have the analogous problem of local minima [23]. Reach-avoid RL gives safety and liveness guarantees by learning a reach-avoid value function [24], a different design from post-hoc filtering.

Safety filters and shields in RL. Shields replace unsafe actions during learning [25, 26], and CBF filters have been combined with RL in several ways [27–30]. Krasowski et al. [12] compare action replacement, projection and masking, and report that a penalty on interventions helps. Thumm et al. [31] modify the agent's action proactively to reduce fail-safe interventions. Pizarro Bejarano et al. [10] study filtering during training, a correction penalty and safe resets for PPO on quadrotor trajectory tracking. Yang et al. [11] filter rollouts and add a CBF reward term so that the policy can be deployed without a filter. Oh et al. [13] prove that a sufficiently permissive filter does not degrade asymptotic optimality. CBF-based safe RL for navigation includes [32–35]. Barriers synthesised online from range data remove the known-map assumption [36, 37].

Evaluation practice. RL results vary strongly between seeds [38], and few-run studies need interval estimates rather than point estimates [39]. Navigation benchmarks such as BARN [40] and Arena-Rosnav [41], and community guidelines [42], stress classical baselines and standardised metrics.

Table 1 positions this work. Entries for other papers reflect the parts we read (see LITERATURE_REVIEW.md for the reading depth); "n.r." means not reported in those parts, not that the paper lacks the property.

3 Problem formulation

Robot and environment. The robot state is $x_t = (p_t, \theta_t)$ with position $p_t \in [0, W]^2$, $W = 10\,\mathrm{m}$, and heading θ_t. The command is $u_t = (v_t, \omega_t) \in \mathcal{U} = [0, v_{\max}] \times [-\omega_{\max}, \omega_{\max}]$ with $v_{\max} = 1\,\mathrm{m/s}$

and $\omega_{\max} = 1\,\mathrm{rad/s}$. The simulator integrates

$$p_{t+1} = p_t + v_t\,\Delta t\,e(\theta_t), \qquad \theta_{t+1} = \theta_t + \omega_t \Delta t, \qquad e(\theta) = (\cos\theta, \sin\theta)^\top, \tag{1}$$

with $\Delta t = 0.1\,\mathrm{s}$. Obstacles are static discs with centres c_i and radii $R_i \in [0.3, 0.8]\,\mathrm{m}$; the robot is a disc of radius $r = 0.3\,\mathrm{m}$.

Barriers and filter. For a safety margin d (0.3 m) and a wall margin m (0.1 m), define

$$\begin{aligned} h_i(p) &= \|p - c_i\| - (R_i + r + d) && \text{(obstacle } i\text{)}, \\ h_W(p) &\in \{p_x - (r+m),\ W - p_x - (r+m),\ p_y - (r+m),\ W - p_y - (r+m)\} && \text{(walls)}. \end{aligned} \tag{2}$$

The safe set is $\mathcal{S} = \{p : h_k(p) \geq 0\ \forall k\}$. The closing rate per unit speed is $a_i(p,\theta) = e(\theta)^\top (c_i - p)/\|c_i - p\|$ for an obstacle, and $-\cos\theta, \cos\theta, -\sin\theta, \sin\theta$ for the four walls. With $\gamma = 0.5\,\mathrm{s}^{-1}$ (so $\gamma\Delta t = 0.05 \leq 1$), the filter returns

$$F(x,u) = \big(\min\{\mathrm{clip}(v), \bar{v}(x)\},\ \mathrm{clip}(\omega)\big), \qquad \bar{v}(x) = \min\Big\{v_{\max},\ \min_{k:\,a_k>0} \frac{\gamma\, h_k(p)}{a_k(p,\theta)}\Big\}. \tag{3}$$

Because every constraint bounds the single variable v, (3) is the exact solution of $\min_v (v - v^{\mathrm{nom}})^2$ subject to $a_k v \leq \gamma h_k$ and $0 \leq v \leq v_{\max}$.

Policies. A policy maps an observation o_t to a nominal command. The observation contains the noisy normalised position, $\sin\theta$ and $\cos\theta$, goal distance and bearing, a proximity feature, and 16 range readings up to 3 m. The closed loop is $x_{t+1} = f(x_t, F(x_t, \pi(o_t)))$. An episode succeeds when the goal is within 0.5 m; it ends with a collision if the robot body touches an obstacle or leaves the workspace; otherwise it times out after 200/300/400 steps (beginner/intermediate/expert).

Definition 1 (Stall classes)**.** A timeout is a *stall* if the net displacement over its last $K = 100$ steps is below $\delta = 0.10\,\mathrm{m}$. A stall is (i) *filter-blocked* (Type I) if the filter reduced the commanded speed in at least half of those steps; (ii) a *self-stop* (Type II) if, in at least half of them, the policy itself commanded a speed of at most $10^{-3}\,\mathrm{m/s}$; (iii) *other* otherwise. A timeout that is not a stall is *moving*.

4 Analysis of the one-step speed filter

Lemma 1 (Forward invariance [14])**.** *If $p_0 \in \mathcal{S}$ and $u_t = F(x_t, \cdot)$ for all t, then $h_k(p_{t+1}) \geq (1 - \gamma\Delta t) h_k(p_t)$ for every barrier k. In particular, $h_k(p_t) > 0$ for all t if $h_k(p_0) > 0$.*

Proof. By convexity of the norm, $\|p_{t+1} - c_i\| \geq \|p_t - c_i\| - \Delta t\, v_t\, a_i(p_t, \theta_t)$. If $a_i > 0$, the filter enforces $v_t a_i \leq \gamma h_i(p_t)$; if $a_i \leq 0$, the barrier cannot decrease. Wall barriers are affine in p, so the same bound holds with equality. Strict positivity follows from $\gamma\Delta t < 1$. □

Proposition 1 (Heading freedom)**.** *Let $\psi : \mathcal{X} \times \mathcal{U} \to [-\omega_{\max}, \omega_{\max}]$ be any map, and let $F_\psi(x,u) = (\min\{\mathrm{clip}(v), \bar{v}(x)\}, \psi(x,u))$. Then Lemma 1 holds for F_ψ.*

Proof. By (1), p_{t+1} depends only on (p_t, θ_t, v_t), and $\bar{v}(x_t)$ depends only on (p_t, θ_t). The turn rate ω_t affects only θ_{t+1}, which the filter evaluates again at step $t+1$. The proof of Lemma 1 therefore applies at every step, whatever ψ returns. □

Remark 1. Proposition 1 has two consequences. First, the one-step filter can never *steer* the robot away from an obstacle: only the policy, or a heading modification ψ, can do so. Second, heading-level remedies for deadlock, such as a tangential turn when the speed limit is active, inherit the safety guarantee without further proof. In continuous time this corresponds to the position barrier having relative degree two with respect to ω [16].

Proposition 2 (Collapse of the admissible headings)**.** *Fix p with $0 \le h_i(p) \le \varepsilon$ for an obstacle i, and a speed $\underline{v} > 0$. Every heading θ with $\bar{v}(p,\theta) \ge \underline{v}$ satisfies $a_i(p,\theta) \le \gamma\varepsilon/\underline{v}$; that is, the angle between $e(\theta)$ and the direction to c_i is at least* $\arccos(\min\{1, \gamma\varepsilon/\underline{v}\})$*. As $\varepsilon \to 0$, this set of headings approaches the closed half-circle $\{\theta : a_i(p,\theta) \le 0\}$.*

Proof. If $a_i(p,\theta) > 0$, then $\bar{v} \le \gamma h_i/a_i \le \gamma\varepsilon/a_i$, and $\bar{v} \ge \underline{v}$ requires $a_i \le \gamma\varepsilon/\underline{v}$. If $a_i \le 0$, the inequality holds trivially. □

Corollary 1 (Stall bound)**.** *If, over K consecutive steps, $h_i(p_t) \le \varepsilon$ and $a_i(p_t,\theta_t) \ge \bar{a} > 0$, then the robot moves at most $K\Delta t\,\gamma\varepsilon/\bar{a}$, whatever speed the policy commands. For a head-on approach ($a_i \equiv 1$) at the speed limit, $h_i(p_{t+1}) = (1-\gamma\Delta t)\,h_i(p_t)$ holds with equality, so the barrier decays geometrically and the robot approaches, but never reaches, the boundary.*

Proof. The displacement per step is $v_t\Delta t \le \bar{v}\Delta t \le \gamma\varepsilon\Delta t/\bar{a}$. For straight motion toward the centre, the convexity bound in Lemma 1 is tight, giving the stated equality. □

Proposition 3 (Self-stops do not depend on the speed limit)**.** *Let F' be any filter whose executed speed satisfies $0 \le v^{\text{exec}} \le \text{clip}(v^{\text{nom}})$. At a state where the deterministic policy commands $\pi_v(o) \le 0$, the executed speed is 0 under F' and without any filter. Hence no change of the speed limit can remove a Type II stall; only a change of the heading (Proposition 1) or of the policy can.*

Proof. $\text{clip}(\pi_v(o)) = 0$, so $0 \le v^{\text{exec}} \le 0$. □

Propositions 2 and 3 give testable predictions. Filter-blocked stalls should occur at $h \approx 0$ with headings within 90° of the obstacle direction. Self-stops should occur at $h > 0$, and they should be unaffected by the filter's speed limit.

5 Experimental setup

Agents. PPO [43] with separate 256×256 tanh actor and critic networks, a diagonal Gaussian policy whose samples are clipped by the environment, generalised advantage estimation and bootstrapping at time-limit truncations. Training ran for 10^6 steps over a three-stage curriculum (beginner, intermediate, expert; advance at 70 % rolling success). Five seeds (0–4) were trained without the filter and five with the filter in the loop. In the filtered runs, the buffer stores the policy's own sample, as in [10]. The checkpoint with the highest success on 60 validation episodes (20 per stage), on seeds disjoint from the test seeds, was kept. One hyperparameter set was used for both variants and was not tuned for the filtered setting. All ten runs reached the expert stage within 43k–76k steps.

Baselines. A go-to-goal heading controller, which ignores obstacles, and a hand-tuned artificial potential field on the range readings [23]. **[EVIDENCE REQUIRED:** Published baselines are missing: correction-penalty PPO [10] (E1), PPO-Lagrangian [44], and a classical local planner such as DWA [45] or MPC with a DT-CBF [15].]

Test scenarios. 100 seeded scenarios per stage (seed bases $2 \cdot 10^6 + 10^5 s$). The expert stage has 8–12 obstacles. A grid oracle (5 cm, 8-connected Dijkstra on the free space inflated by $r + d$) found a margin-keeping path in all 300 scenarios. Two properties of the generator must be kept in mind:

- The goal is clipped to $[1.5, 8.5]^2$, so 53 of the 100 expert goals are closer than the configured minimum of 5 m (maximum 8.9 m).
- Three scenarios (two beginner, one expert) start inside the goal region and count as successes at step 1 for every controller.

The 0.3 m margin and the sampled-action evaluation were chosen after inspecting these test scenarios, so the test set is not a clean held-out set for those two choices. **[EVIDENCE REQUIRED:** Re-evaluation on fresh scenarios from a corrected generator (E3).]

Conditions and classification. Each PPO agent was evaluated with the filter off and on, using the policy mean. Each was additionally evaluated with the filter on and actions sampled from the policy, seeded per episode. Every timeout with the filter on was replayed deterministically from its seed and classified by Definition 1. This classification is a post-hoc analysis made after the benchmark had been run.

Statistics. For PPO, the unit of replication is a training seed. Seed-level 95 % t-intervals over five seeds describe variation between training runs on the fixed scenario set. Comparisons between conditions use a paired hierarchical bootstrap (10,000 replicates). Scenarios are resampled jointly for both conditions, because all conditions share the same scenarios, and seeds are resampled within each condition. Proportions for the deterministic baselines use Wilson intervals over episodes. A two-way decomposition of episode success into seed, scenario and residual sums of squares quantifies the two sources of uncertainty. No multiplicity correction was applied, because the contrasts are reported as estimates with intervals and are not used as a family of hypothesis tests. With five seeds, the bootstrap underestimates seed variance, so the intervals should be read as approximate.

Reproducibility. The repository contains the committed checkpoints, and regenerating the benchmark from them reproduces the per-episode files byte for byte. All analysis scripts read only stored outputs; Section 9 lists them.

6 Results

6.1 Safety and success

With the filter on, none of the 3,600 test episodes (all controllers and stages) had a collision, and the smallest obstacle clearance was the 0.30 m margin (Table 2). The filter's cost in success depends on the controller. For PPO trained without the filter, expert success fell from 94.6 % to 65.0 % when the filter was added at test time, a paired difference of 29.6 points [21.6, 38.4]. The agent trained with the filter reached 61.0 %. The paired difference to the unfiltered-trained agent with the filter on was −4.0 points [−10.8, 3.2] with mean actions, and +6.4 points [−1.6, 15.2] with sampled actions. Neither comparison shows that training with the filter improves success.

6.2 Two stall mechanisms

Figure 1 and Table 3 classify every timeout with the filter on.

Table 2: Expert stage, 100 test scenarios. PPO: mean over five training seeds with 95 % t-interval over seeds (500 episodes); baselines: Wilson interval over 100 episodes. Sources: `results/benchmark/summary.csv`, `results/stochastic_eval/summary.json`.

Condition	Success (%)	Collision (%)	Timeout (%)	Interventions (%)
PPO trained w/o filter, filter off	94.6 [91.7, 97.5]	1.0	4.4	–
PPO trained w/o filter, filter on	65.0 [61.5, 68.5]	0	35.0	58.0
PPO trained w/o filter, filter on, sampled	70.0 [66.3, 73.7]	0	30.0	n.r.
PPO trained with filter, filter on	61.0 [55.4, 66.6]	0	39.0	41.5
PPO trained with filter, filter on, sampled	76.4 [68.2, 84.6]	0	23.6	n.r.
Go-to-goal, filter off	57.0 [47.2, 66.3]	43.0	0	–
Go-to-goal, filter on	46.0 [36.6, 55.7]	0	54.0	67.2
Potential field, filter off	73.0 [63.6, 80.7]	0	27.0	–
Potential field, filter on	72.0 [62.5, 79.9]	0	28.0	40.7

Table 3: Timeout classes with the filter on (counts; PPO over 500 episodes per row, baselines over 100). Source: `research/audit_outputs/stall_taxonomy_all.csv`.

	Intermediate			Expert		
Condition	Type I	Type II	moving	Type I	Type II	moving
PPO trained w/o filter, mean	68	13	0	141	34	0
PPO trained w/o filter, sampled	66	0	4	147	1	2
PPO trained with filter, mean	28	68	4	67	122	6
PPO trained with filter, sampled	20	8	32	71	13	34
Go-to-goal	39	0	0	54	0	0
Potential field	0	4	7	0	6	22

- **PPO trained without the filter.** Filter-blocked stalls dominate: 141 of 175 expert timeouts and 68 of 81 intermediate timeouts, with mean actions. In these stalls the policy commands full speed (median 1.0 m/s) while the turn rate changes sign a median of 70 times in 100 steps.
- **PPO trained with the filter.** Self-stops dominate: 122 of 195 expert timeouts and 68 of 100 intermediate timeouts. The policy commands zero speed at a median barrier value of 0.41 m (interquartile range 0.16–0.72 m), well before the keep-out boundary. In 89 of the 122 expert self-stops, the goal lies within 45° of the direction of the nearest obstacle (median 26°), and the robot barely turns (median $|\omega|$ 0.04 rad/s).
- **Baselines.** The two classical controllers bracket the PPO behaviour. Go-to-goal, which always commands forward speed toward the goal, fails only by filter-blocking (54 of 54 expert timeouts). The potential field never stalls against the filter: it oscillates in local minima (22 "moving" timeouts) or stops (6).

The taxonomy is insensitive to its thresholds. Varying the displacement threshold (0.05, 0.10 and 0.20 m) and the share threshold (30, 50 and 70 %) changes the expert counts by at most 3 (filter-blocked) and 3 (self-stop) for either agent type. Varying the window length K over 50, 100, 150 and 200 steps, with the displacement threshold scaled to the window, leaves the counts of the agent trained without the filter unchanged (141 and 34). For the agent trained with the filter, the filter-blocked count ranges over 63–69 and the self-stop count over 119–122.

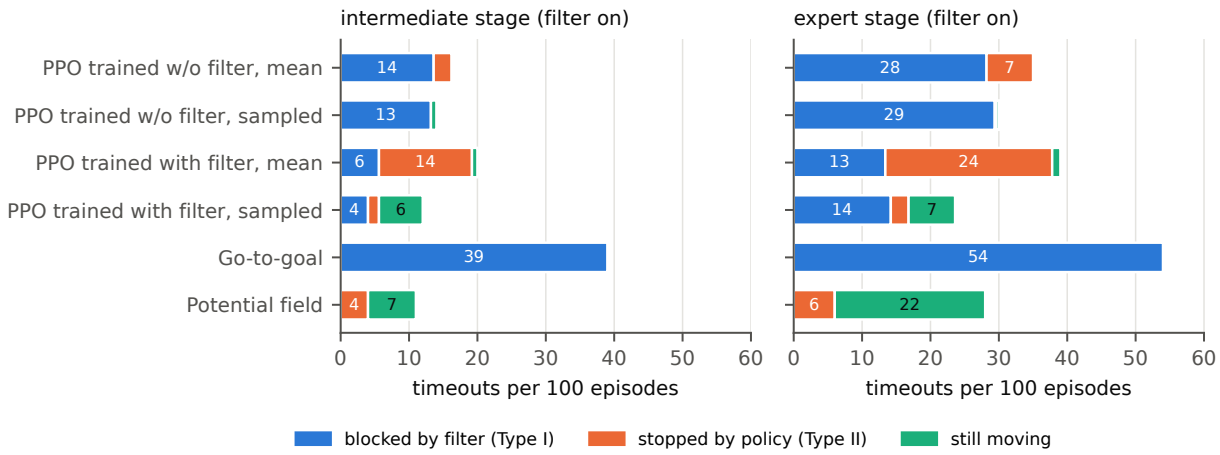


Figure 1: Timeouts per 100 episodes with the filter on, by stall class (Definition 1). PPO: five seeds × 100 scenarios; baselines: 100 scenarios. Source: `research/audit_outputs/stall_taxonomy_all.csv`.

6.3 The geometry of filter-blocked stalls matches the analysis

Proposition 2 predicts that a robot blocked at the boundary heads less than 90° from the obstacle direction. The data confirm this (Figure 2a).

- 617 of the 618 filter-blocked stalls of all PPO conditions have a heading below 90°; the remaining one is at 91.3°. The median is 66.8°, so the robots are typically not facing the obstacle head-on.
- The obstacle barrier at the end of these stalls is at most 4.2 mm, consistent with the geometric decay in Corollary 1.
- In all 618 stalls, the active barrier is an obstacle: no wall barrier is below 1 mm.

Self-stops, in contrast, are spread over barrier values from 0 to 1.4 m (Figure 2b). This is consistent with Proposition 3: these stops are a property of the policy, not of the filter.

6.4 Training with the filter shifts the mechanism; sampling removes self-stops

Paired comparisons (Figure 3) quantify both effects.

- **Training with the filter.** On the expert stage, it lowered the filter-blocked rate by 14.8 points [5.6, 24.4] and raised the self-stop rate by 17.6 points [8.2, 27.0]. On the intermediate stage, the corresponding changes were −8.0 [−14.8, −1.8] and +11.0 [5.6, 17.2].
- **Sampled actions** (agent trained with the filter). The expert self-stop rate fell by 21.8 points [12.4, 31.6], while the filter-blocked rate did not change (+0.8 [−7.6, 10.0]).
- **Success.** Sampling raised expert success by 15.4 points [7.8, 24.0] for the filter-trained agent, against 5.0 [0.4, 10.4] for the agent trained without the filter.

Stochastic deployment therefore acts on the learned mechanism only. The filter-blocked stalls persist, as Proposition 2 suggests they should for any policy that keeps a heading within 90° of the obstacle.

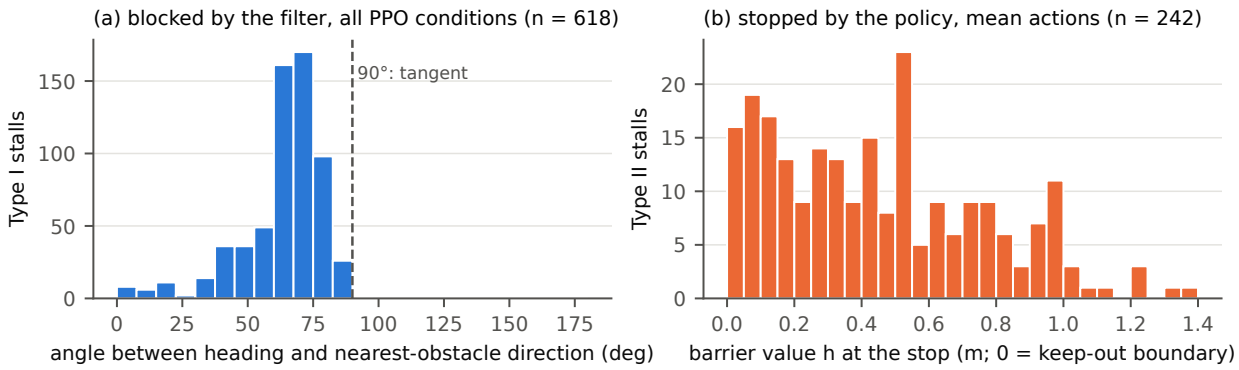


Figure 2: (a) Heading relative to the nearest obstacle at the end of filter-blocked stalls (all PPO conditions, both stages); the dashed line marks 90° (tangent). (b) Barrier value at self-stops (PPO, mean actions). Source: `research/audit_outputs/stall_taxonomy_all.csv`.

6.5 Sources of uncertainty

The decomposition of expert success shows how much the test scenario matters. Depending on the condition, the scenario term accounts for 25–86 % of the sum of squares, against under 2 % for the seed term. Seed-only intervals such as those in Table 2 therefore describe reproducibility of training on these 100 scenarios; they understate the uncertainty about the scenario distribution. An approximate fit of interval width to the number of seeds and scenarios suggests that about 10 seeds and 1,000 scenarios per stage would give half-widths near 4–5 points for differences of this kind. Beyond about 1,000 scenarios, more seeds help more than more scenarios.

6.6 Remedies

[**EVIDENCE REQUIRED:** E1: correction-penalty PPO [10] with $\alpha = 1$ against an identical control (five seeds each), evaluated on fresh scenarios. The decisive quantity is the change in the expert deadlock rate by class.] [**EVIDENCE REQUIRED:** E2: a heading-level escape modification ψ (safe by Proposition 1) applied at test time to the ten frozen agents, with Type I and Type II rates on fresh scenarios.]

7 Discussion

Two different problems. Most discussions of filter-induced deadlock treat it as one problem of the filter [5, 6]. Our data show two different problems, and which one dominates depends on the training setup. A policy that never saw the filter keeps pushing at the boundary. It is held there because, by Proposition 2, no heading within 90° of the obstacle admits motion. This is the learned-policy counterpart of the undesired boundary equilibria of CBF-QPs. A policy trained with the filter mostly avoids being blocked, but it has learned to stop on its own when an obstacle lies between it and the goal. By Proposition 3, no change of the speed limit can affect such stops.

Why do filter-trained policies stop? We have no causal evidence yet. A plausible mechanism is *action aliasing* in the filtered environment. Every sampled speed $v \leq 0$ is executed as 0, and every speed above the limit is executed as the limit. Within these sets, the executed action, and therefore the return, does not depend on the sampled value, so the policy-gradient signal cannot

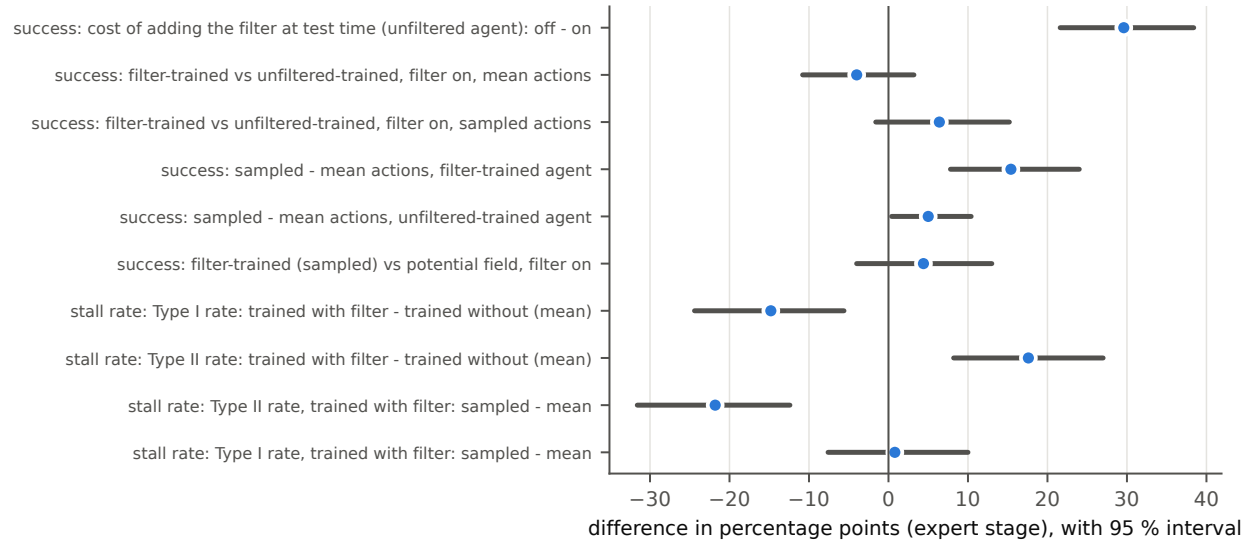


Figure 3: Paired differences (expert stage) with 95 % paired hierarchical bootstrap intervals. Success differences: `research/audit_outputs/paired_comparisons_expert.csv`; stall-rate differences: `paper_draft/generated/stats.json`.

distinguish these actions. The mean can then drift into the stopping region near obstacles, where forward commands were often truncated during training. Sampled actions occasionally leave this region, which is consistent with the observed reduction of self-stops. The non-potential-based risk penalty in the reward is an alternative or additional explanation.

Hypothesis 1. Removing the aliasing (a bounded action distribution, or a policy parameterised in the executed-action space) removes Type II stops, while a reward-only change does not. **[EVIDENCE REQUIRED:** Untested; minimum_experiments.md, E4.]

Relation to theory on filtered training. Oh et al. [13] show that a sufficiently permissive filter does not limit the asymptotically optimal return. The speed filter here permits turning in place, and the oracle shows that a margin-keeping path exists in every test scenario. A large part of the remaining gap is therefore plausibly a learning problem rather than a limit imposed by the filter. Checking the formal permissiveness condition for this filter is left open.

Implications for remedy design. Proposition 1 means that heading-level remedies come at no cost to the one-step guarantee. Proposition 3 means that any remedy acting only on the speed limit (for example a less conservative γ) cannot remove self-stops. The published correction penalty [10] penalises both the filter's truncation and the box clipping of $v \leq 0$. It may therefore act on both mechanisms, which makes it the most important missing baseline.

8 Limitations and threats to validity

- **External validity.** 2D simulation with exact Euler kinematics, static circular obstacles, exact state for the filter and a known map. Nothing was tested on a robot, and no finite simulation establishes real-world safety. The filter's guarantee holds at the sampled states. Between

samples, the chord bound limits the penetration to about 1.4 mm for the generated obstacle sizes; continuous-time motion is not covered.

- **Scenario design.** Goal clipping shortens many goal distances. Three trivial scenarios add 1–2 points to every success rate in their stages. The layouts are fairly open: the median oracle path is 6–8 % longer than the straight line.
- **Test-set reuse.** The margin, the sampled-action evaluation and the stall analysis were all developed on the same test scenarios.
- **Statistical power.** Five seeds per variant and 100 scenarios per stage. The intervals are approximate, and there is no multiplicity correction.
- **Construct validity of the taxonomy.** The class thresholds are heuristic, but varying the displacement threshold, the share threshold and the window length changed the expert counts by at most 4.
- **Baselines and tuning.** PPO hyperparameters were not tuned for the filtered setting, the potential field was tuned by hand, and published safe-RL and classical baselines are missing.
- **Privileged feature.** The proximity feature in the observation is computed from the true map.

9 Conclusion

A one-step discrete-time CBF filter makes learned navigation policies collision-free in simulation, but it does not make them live. We showed that the filter's one-step guarantee is independent of the turn rate, and that the headings admitting motion collapse to a half-plane at the boundary. We also showed that learned policies deadlock behind the filter by two distinct mechanisms. A policy trained without the filter is blocked by the filter at the boundary, while a policy trained with the filter mostly stops itself before it. Stochastic deployment removes the second mechanism but not the first. These results separate what a filter design can fix from what training must fix. **[EVIDENCE REQUIRED:** The concluding claim about which remedy resolves which mechanism depends on E1 and E2.]

Data and code availability

Code, committed checkpoints, per-episode results and analysis scripts: `https://github.com/pouyapd/SafeNav-RL`. Benchmark: `results/benchmark/`; sampled-action evaluation: `results/stochastic_eval/`; stall taxonomy: `research/scripts/stall_taxonomy.py`; figures and statistics in this paper: `research/scripts/paper_figures_stats.py`. **[EVIDENCE REQUIRED:** Archive a tagged release with a DOI (e.g. Zenodo) before submission.]

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